

House Price Analysis & Prediction

An end-to-end Data Analysis and Machine Learning project that analyzes housing data and predicts house prices based on key features such as bedrooms, bathrooms, house size, and location (zip code) using Python.

Project Overview

The goal of this project is to:

Perform data cleaning and exploratory data analysis (EDA)

Understand factors affecting house prices

Build a machine learning model to predict house prices accurately

Follow a production-style ML pipeline

This project simulates a real-world real estate price prediction workflow.

Dataset Information

Total Records: 2016

Features:

beds – Number of bedrooms

baths – Number of bathrooms

size – House size in square feet

zip_code – Location of the house

price – House price (Target variable)

Data Cleaning & Feature Engineering

The following preprocessing steps were applied:

Removed unnecessary columns

Checked and handled missing values

Converted zip_code to a categorical feature

Created a new feature: price_per_sqft

Applied logical data cleaning:

Removed unrealistic house sizes per bedroom

Removed outliers using zip-code-wise standard deviation

Removed illogical bedroom–bathroom combinations

✅ Final clean dataset size: 1494 records

📊 Exploratory Data Analysis (EDA)

EDA was performed using Matplotlib and Seaborn:

Price vs House Size (scatter plot)

House price distribution (histogram + KDE)

Price variation by number of bedrooms (boxplot)

Bedroom vs bathroom distribution (count plot)

These visualizations helped identify strong relationships between house size, location, and price.

🤖 Machine Learning Model

Model Used: Linear Regression

Techniques:

One-Hot Encoding for categorical variables

Feature scaling using StandardScaler

Pipeline and ColumnTransformer for clean workflow

Train-Test Split: 80% training / 20% testing

The model was trained and tested successfully using a production-ready pipeline.

Technologies Used

Python

Pandas

NumPy

Matplotlib

Seaborn

Scikit-learn

Jupyter Notebook

Key Learnings

Real-world data cleaning and preprocessing

Outlier detection and removal

Feature engineering techniques

Data visualization for insights

Building scalable machine learning pipelines

Project Outcome

This project enhanced my understanding of data analysis, visualization, and machine learning, and provided hands-on experience with real-world housing data and predictive modeling.

How to Run the Project

Install required libraries

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

Open the Jupyter Notebook

```
jupyter notebook
```

 Author

Sameer Ali

Aspiring Data Analyst / Data Scientist

 LinkedIn: (<https://linkedin.com/in/sameer-ali-028ab13a1>)

 GitHub: (<https://github.com/sameerali077>)